Fair Classifier and HYB-MS GAN Augmentation Study

Same 363,692 real URLs • Same 35 PSL features where applicable
Seeds 42/43/44 • Frozen external 10K • MS-HardGAN-C

Purpose: remove feature-access favoritism, isolate feature fusion and adaptation, and test whether GAN augmentation improves each unchanged model.

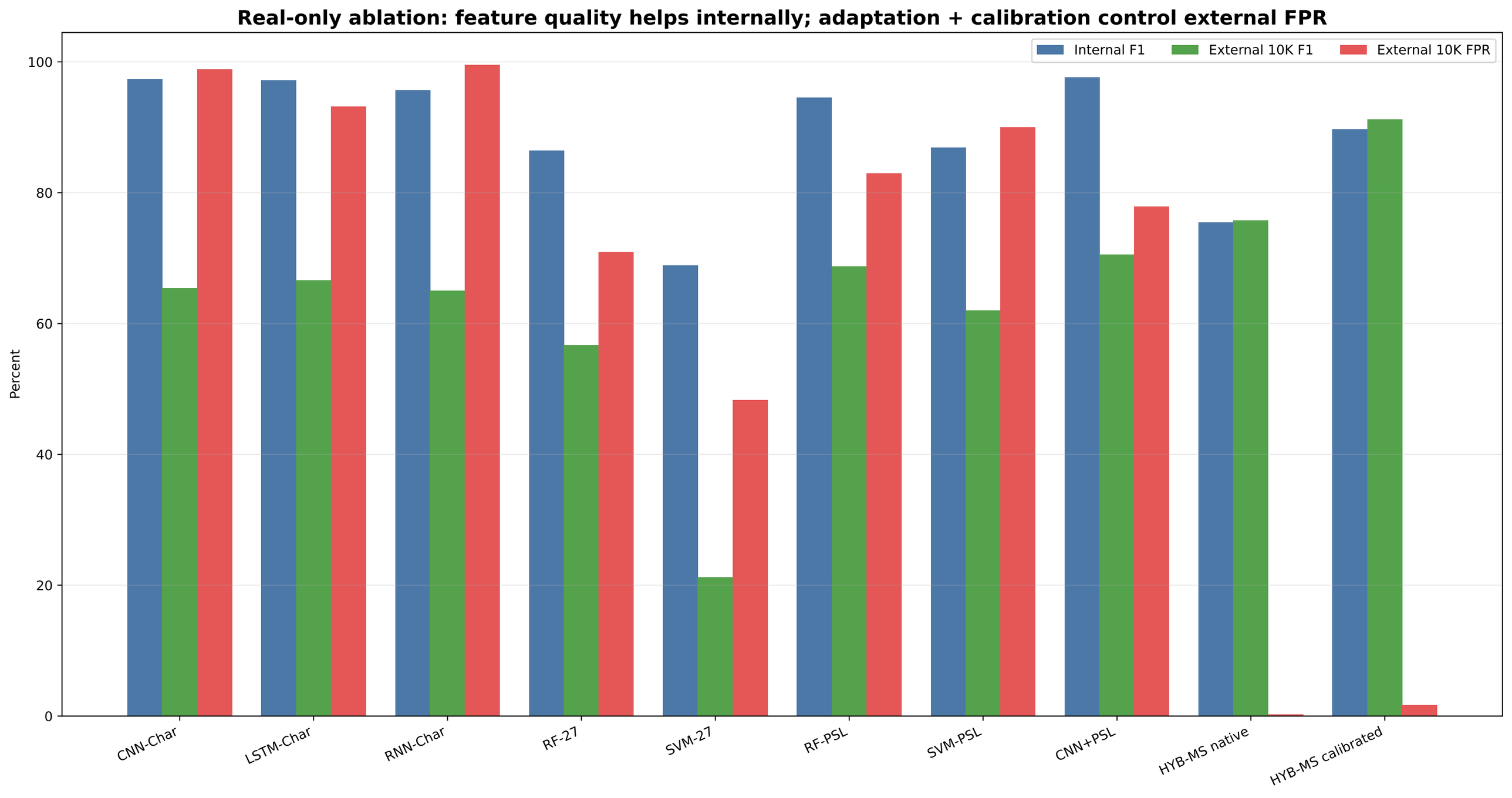
Main result: 35 PSL-aware features substantially strengthen RF/SVM internally but do not solve external source shift. The separately calibrated HYB-MS +10% ensemble is the strongest balanced augmented candidate on the frozen external 10K.

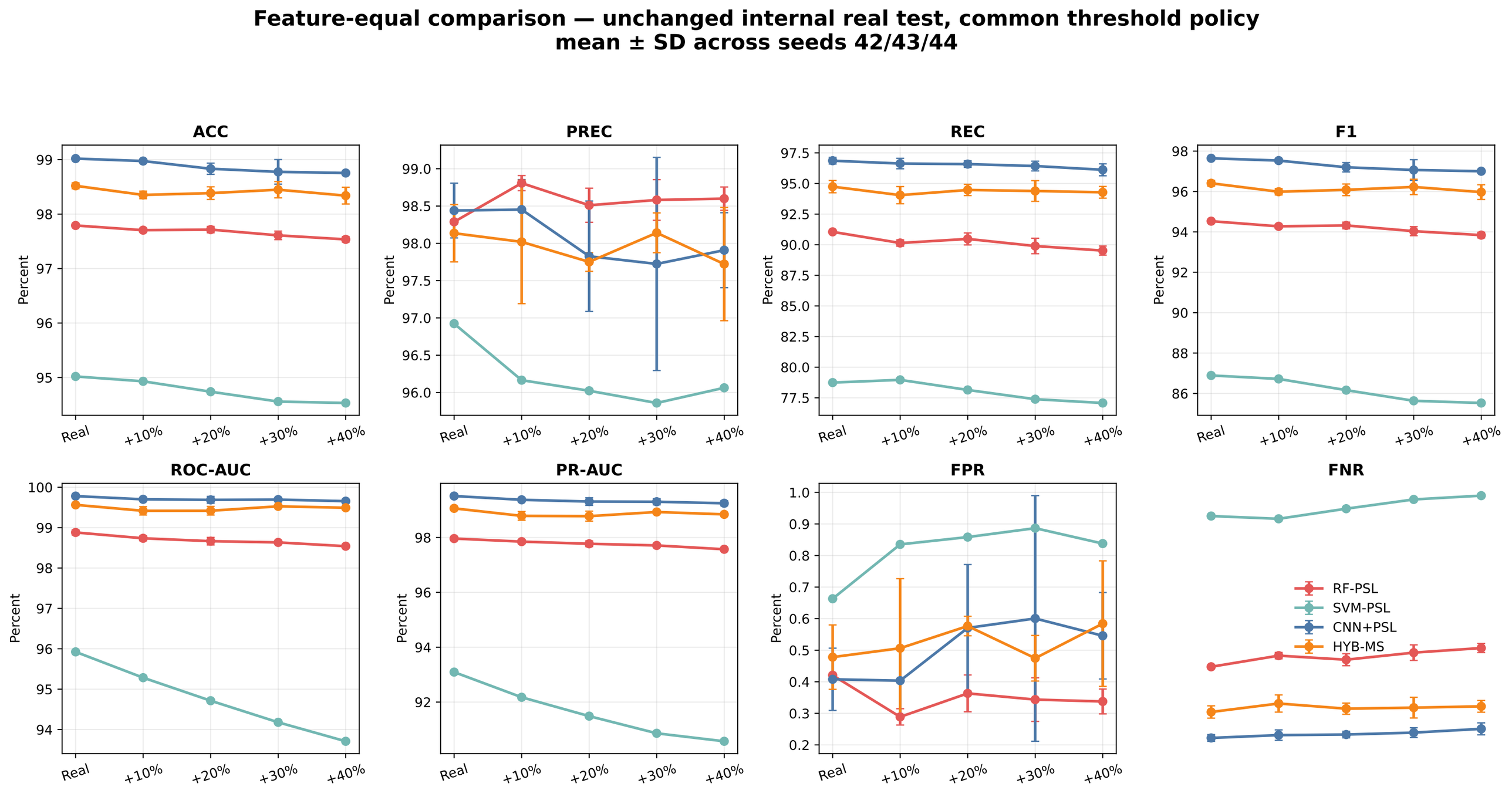
Scientific guard: strong metrics mean deployment candidate—not deployment-ready. A later temporal test, realistic prevalence, browser latency/memory, drift monitoring, and operational false-alert testing remain required.

What is controlled—and what is intentionally different

System	Real rows	Feature view	Training distinction	Threshold source
CNN/LSTM/RNN-Char	363,692	128 chars	Ordinary training	Original validation
RF-27 / SVM-27	363,692	27 lexical	Class weighting	Original validation
RF-PSL / SVM-PSL	363,692	Same 35 PSL	Class weighting	Original validation
CNN+PSL	363,692	Chars + same 35 PSL	Ordinary training	Original validation
HYB-MS	363,692	Chars + same 35 PSL	Source balance + CORAL	Original + domain val.
HYB-MS calibrated	363,692	Same as HYB-MS	3-seed ensemble	Separate 400-row cal.

Fairness rule: RF-PSL, SVM-PSL, CNN+PSL, and HYB-MS receive the exact same 35 robust-scaled PSL features. HYB-MS balancing/CORAL are reported as explicit method components, not hidden advantages. Synthetic rows are a declared group and are excluded from CORAL pairs.


Real-only ablation: feature quality helps internally; adaptation + calibration control external FPR
Internal F1
External 10K F1
External 10K FPR
Percent
0
20
40
60
80
100
CNN-Char
LSTM-Char
RNN-Char
RF-27
SVM-27
RF-PSL
SVM-PSL
CNN+PSL
HYB-MS native
HYB-MS calibrated


Feature-equal comparison — unchanged internal real test, common threshold policy
mean ± SD across seeds 42/43/44
ACC
PREC
REC
F1
ROC-AUC
PR-AUC
FPR
FNR
Percent
Real
+10%
+20%
+30%
+40%
RF-PSL
SVM-PSL
CNN+PSL
HYB-MS

Internal real test — common original-validation threshold (rows 1–10)

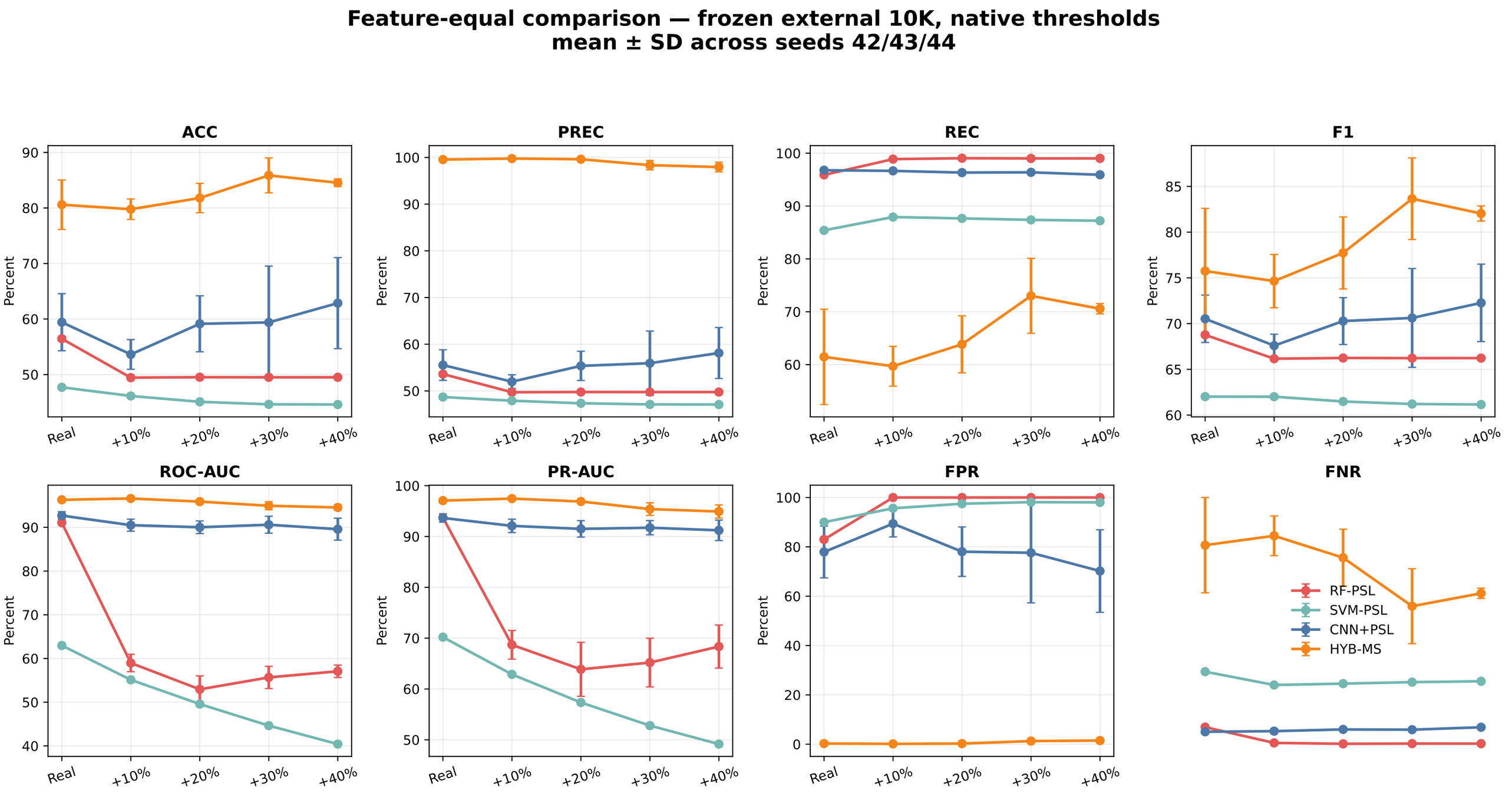
Model	Condition	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR
RF-PSL	Real	0.9779±0.0003	0.9829±0.0003	0.9105±0.0012	0.9453±0.0007	0.9888±0.0006	0.9796±0.0006	0.0042±0.0001	0.0895±0.0012
RF-PSL	+10%	0.9770±0.0003	0.9881±0.0010	0.9014±0.0026	0.9427±0.0010	0.9874±0.0006	0.9785±0.0007	0.0029±0.0003	0.0986±0.0026
RF-PSL	+20%	0.9772±0.0006	0.9851±0.0023	0.9047±0.0049	0.9432±0.0016	0.9866±0.0009	0.9777±0.0010	0.0036±0.0006	0.0953±0.0049
RF-PSL	+30%	0.9761±0.0008	0.9858±0.0027	0.8989±0.0063	0.9404±0.0022	0.9863±0.0001	0.9771±0.0001	0.0034±0.0007	0.1011±0.0063
RF-PSL	+40%	0.9754±0.0005	0.9860±0.0016	0.8952±0.0037	0.9384±0.0014	0.9854±0.0002	0.9757±0.0001	0.0034±0.0004	0.1048±0.0037
SVM-PSL	Real	0.9502±0.0000	0.9692±0.0000	0.7874±0.0000	0.8689±0.0000	0.9592±0.0000	0.9309±0.0000	0.0066±0.0000	0.2126±0.0000
SVM-PSL	+10%	0.9493±0.0000	0.9617±0.0000	0.7897±0.0000	0.8672±0.0000	0.9528±0.0000	0.9218±0.0000	0.0084±0.0000	0.2103±0.0000
SVM-PSL	+20%	0.9474±0.0000	0.9602±0.0000	0.7814±0.0000	0.8616±0.0000	0.9471±0.0000	0.9149±0.0000	0.0086±0.0000	0.2186±0.0000
SVM-PSL	+30%	0.9456±0.0000	0.9586±0.0000	0.7739±0.0000	0.8564±0.0000	0.9418±0.0000	0.9086±0.0000	0.0089±0.0000	0.2261±0.0000
SVM-PSL	+40%	0.9453±0.0000	0.9606±0.0000	0.7708±0.0000	0.8553±0.0000	0.9371±0.0000	0.9057±0.0000	0.0084±0.0000	0.2292±0.0000

All values are mean ± sample SD across three seeds. FPR/FNR are threshold-dependent; ROC-AUC/PR-AUC evaluate ranking.

Internal real test — common original-validation threshold (rows 11–20)

Model	Condition	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR
CNN+PSL	Real	0.9902±0.0002	0.9844±0.0037	0.9686±0.0027	0.9764±0.0005	0.9978±0.0001	0.9951±0.0001	0.0041±0.0010	0.0314±0.0027
CNN+PSL	+10%	0.9897±0.0003	0.9845±0.0040	0.9663±0.0042	0.9753±0.0007	0.9970±0.0002	0.9937±0.0003	0.0040±0.0011	0.0337±0.0042
CNN+PSL	+20%	0.9883±0.0010	0.9783±0.0074	0.9658±0.0027	0.9720±0.0023	0.9969±0.0008	0.9931±0.0014	0.0057±0.0020	0.0342±0.0027
CNN+PSL	+30%	0.9878±0.0023	0.9772±0.0143	0.9642±0.0040	0.9706±0.0051	0.9969±0.0004	0.9930±0.0011	0.0060±0.0039	0.0358±0.0040
CNN+PSL	+40%	0.9875±0.0001	0.9791±0.0050	0.9612±0.0049	0.9700±0.0003	0.9966±0.0004	0.9925±0.0009	0.0055±0.0014	0.0388±0.0049
HYB-MS	Real	0.9852±0.0005	0.9814±0.0038	0.9474±0.0050	0.9641±0.0014	0.9956±0.0001	0.9906±0.0003	0.0048±0.0010	0.0526±0.0050
HYB-MS	+10%	0.9835±0.0007	0.9802±0.0083	0.9405±0.0070	0.9599±0.0016	0.9942±0.0010	0.9879±0.0016	0.0051±0.0022	0.0595±0.0070
HYB-MS	+20%	0.9839±0.0012	0.9775±0.0013	0.9447±0.0046	0.9608±0.0029	0.9942±0.0011	0.9878±0.0018	0.0058±0.0003	0.0553±0.0046
HYB-MS	+30%	0.9845±0.0015	0.9814±0.0027	0.9439±0.0084	0.9623±0.0038	0.9953±0.0002	0.9893±0.0005	0.0047±0.0007	0.0561±0.0084
HYB-MS	+40%	0.9834±0.0015	0.9772±0.0076	0.9428±0.0048	0.9597±0.0036	0.9949±0.0001	0.9884±0.0002	0.0058±0.0020	0.0572±0.0048

All values are mean ± sample SD across three seeds. FPR/FNR are threshold-dependent; ROC-AUC/PR-AUC evaluate ranking.


Feature-equal comparison — frozen external 10K, native thresholds
mean ± SD across seeds 42/43/44
ACC
PREC
REC
F1
ROC-AUC
PR-AUC
FPR
FNR
Percent
Real
+10%
+20%
+30%
+40%
RF-PSL
SVM-PSL
CNN+PSL
HYB-MS

Frozen external 10K — native threshold policy (rows 1–10)

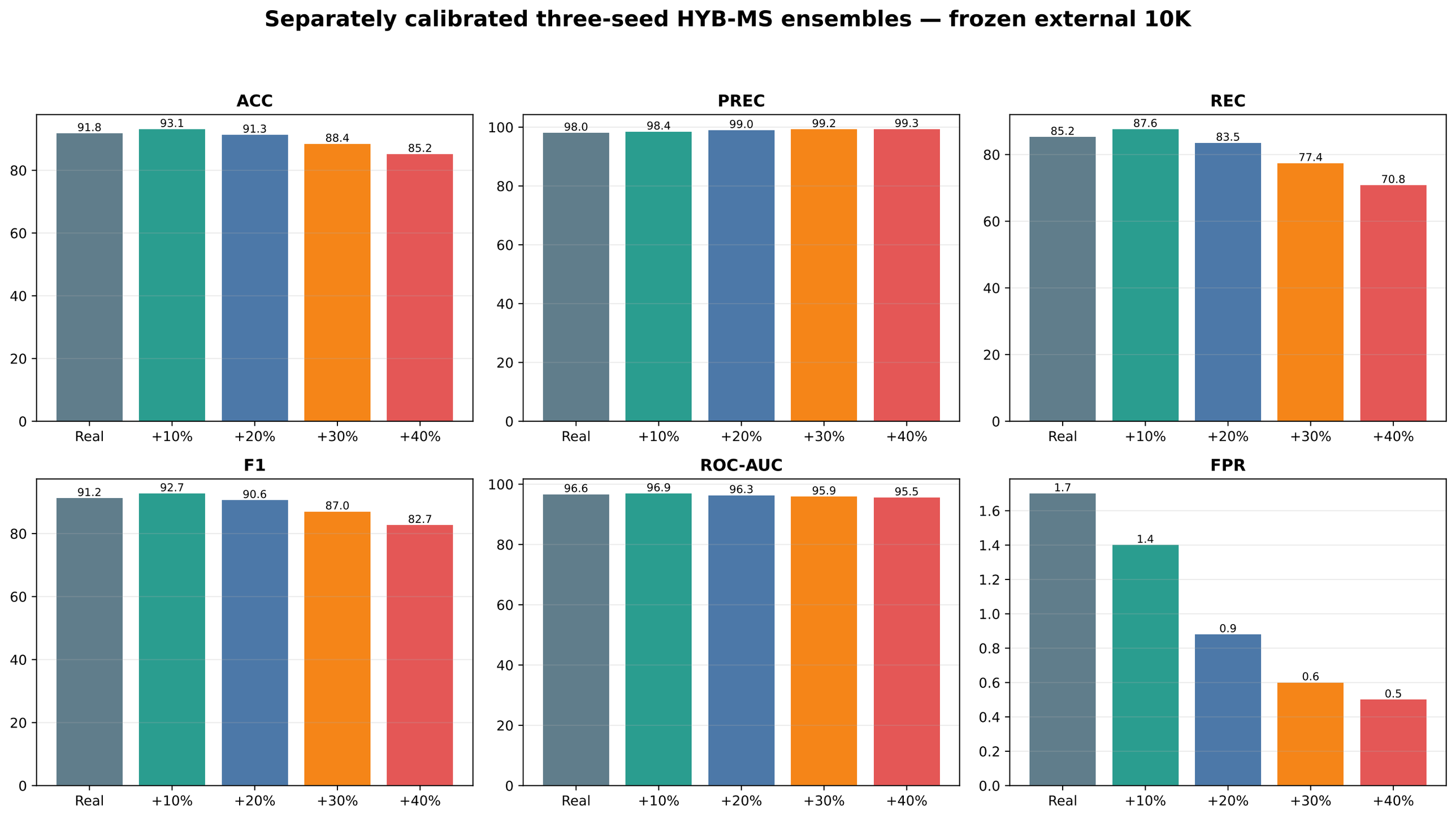
Model	Condition	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR
RF-PSL	Real	0.5645±0.0034	0.5361±0.0020	0.9589±0.0004	0.6877±0.0016	0.9106±0.0004	0.9373±0.0005	0.8299±0.0070	0.0411±0.0004
RF-PSL	+10%	0.4944±0.0004	0.4972±0.0002	0.9887±0.0008	0.6616±0.0004	0.5897±0.0200	0.6869±0.0283	1.0000±0.0000	0.0113±0.0008
RF-PSL	+20%	0.4952±0.0003	0.4976±0.0001	0.9905±0.0005	0.6624±0.0002	0.5294±0.0308	0.6387±0.0532	1.0000±0.0000	0.0095±0.0005
RF-PSL	+30%	0.4950±0.0001	0.4975±0.0001	0.9901±0.0002	0.6622±0.0001	0.5567±0.0255	0.6520±0.0480	1.0000±0.0000	0.0099±0.0002
RF-PSL	+40%	0.4951±0.0002	0.4975±0.0001	0.9901±0.0004	0.6623±0.0002	0.5707±0.0143	0.6835±0.0424	1.0000±0.0000	0.0099±0.0004
SVM-PSL	Real	0.4770±0.0000	0.4869±0.0000	0.8540±0.0000	0.6202±0.0000	0.6297±0.0000	0.7020±0.0000	0.9000±0.0000	0.1460±0.0000
SVM-PSL	+10%	0.4614±0.0000	0.4790±0.0000	0.8792±0.0000	0.6201±0.0000	0.5512±0.0000	0.6288±0.0000	0.9564±0.0000	0.1208±0.0000
SVM-PSL	+20%	0.4509±0.0000	0.4735±0.0000	0.8766±0.0000	0.6149±0.0000	0.4958±0.0000	0.5734±0.0000	0.9748±0.0000	0.1234±0.0000
SVM-PSL	+30%	0.4463±0.0000	0.4711±0.0000	0.8738±0.0000	0.6121±0.0000	0.4464±0.0000	0.5279±0.0000	0.9812±0.0000	0.1262±0.0000
SVM-PSL	+40%	0.4460±0.0000	0.4708±0.0000	0.8722±0.0000	0.6116±0.0000	0.4042±0.0000	0.4916±0.0000	0.9802±0.0000	0.1278±0.0000

All values are mean ± sample SD across three seeds. FPR/FNR are threshold-dependent; ROC-AUC/PR-AUC evaluate ranking.

Frozen external 10K — native threshold policy (rows 11–20)

Model	Condition	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR
CNN+PSL	Real	0.5943±0.0514	0.5553±0.0328	0.9677±0.0034	0.7053±0.0259	0.9269±0.0088	0.9365±0.0082	0.7792±0.1049	0.0323±0.0034
CNN+PSL	+10%	0.5362±0.0268	0.5198±0.0147	0.9665±0.0036	0.6759±0.0125	0.9050±0.0138	0.9210±0.0132	0.8941±0.0538	0.0335±0.0036
CNN+PSL	+20%	0.5914±0.0504	0.5536±0.0313	0.9633±0.0008	0.7028±0.0256	0.9002±0.0144	0.9151±0.0162	0.7805±0.1002	0.0367±0.0008
CNN+PSL	+30%	0.5939±0.1015	0.5593±0.0689	0.9637±0.0025	0.7062±0.0540	0.9061±0.0194	0.9174±0.0140	0.7760±0.2031	0.0363±0.0025
CNN+PSL	+40%	0.6286±0.0820	0.5812±0.0546	0.9591±0.0036	0.7227±0.0424	0.8958±0.0252	0.9122±0.0201	0.7019±0.1674	0.0409±0.0036
HYB-MS	Real	0.8059±0.0445	0.9956±0.0016	0.6147±0.0901	0.7575±0.0685	0.9631±0.0005	0.9707±0.0007	0.0028±0.0012	0.3853±0.0901
HYB-MS	+10%	0.7978±0.0185	0.9977±0.0013	0.5969±0.0376	0.7465±0.0291	0.9661±0.0022	0.9746±0.0021	0.0014±0.0009	0.4031±0.0376
HYB-MS	+20%	0.8180±0.0265	0.9962±0.0018	0.6384±0.0539	0.7772±0.0394	0.9589±0.0035	0.9689±0.0023	0.0025±0.0013	0.3616±0.0539
HYB-MS	+30%	0.8587±0.0314	0.9836±0.0099	0.7301±0.0709	0.8365±0.0446	0.9494±0.0090	0.9540±0.0124	0.0127±0.0083	0.2699±0.0709
HYB-MS	+40%	0.8455±0.0068	0.9795±0.0102	0.7057±0.0097	0.8204±0.0083	0.9455±0.0061	0.9492±0.0129	0.0148±0.0075	0.2943±0.0097

All values are mean ± sample SD across three seeds. FPR/FNR are threshold-dependent; ROC-AUC/PR-AUC evaluate ranking.


Separately calibrated three-seed HYB-MS ensembles — frozen external 10K
ACC
91.8
93.1
91.3
88.4
85.2
PREC
98.0
98.4
99.0
99.2
99.3
REC
85.2
87.6
83.5
77.4
70.8
F1
91.2
92.7
90.6
87.0
82.7
ROC-AUC
96.6
96.9
96.3
95.9
95.5
FPR
1.7
1.4
0.9
0.6
0.5
Real
+10%
+20%
+30%
+40%

Calibrated ensemble: complete frozen-test metrics

Condition	Threshold	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR	GAN REC
Real	0.987956	0.9177	0.9804	0.8524	0.9120	0.9663	0.9749	0.0170	0.1476	0.6309
+10%	0.989095	0.9310	0.9843	0.8760	0.9270	0.9689	0.9781	0.0140	0.1240	0.8490
+20%	0.996419	0.9130	0.9896	0.8348	0.9056	0.9630	0.9744	0.0088	0.1652	0.8531
+30%	0.999349	0.8839	0.9923	0.7738	0.8695	0.9585	0.9691	0.0060	0.2262	0.8455
+40%	0.998698	0.8515	0.9930	0.7080	0.8266	0.9555	0.9664	0.0050	0.2920	0.8509

Best balanced condition: HYB-MS +10%. It improves external F1, accuracy, precision, recall, ROC-AUC and challenge recall relative to calibrated real-only while reducing FPR. Ratios 20–40% progressively sacrifice recall/F1 despite lower FPR.

Deployment-readiness decision

Gate	Status	Evidence / remaining work
Frozen balanced external 10K	PASS as candidate	F1 0.927, ROC-AUC 0.969, FPR 1.40%
Independent threshold calibration	PASS	Separate frozen 400-row calibration set
Later untouched temporal confirma	NOT PASSED	Required before deployment claim
Real browser prevalence / alert bur	NOT TESTED	Balanced 50/50 test is not browsing prevalence
Extension latency, memory, packag	NOT TESTED	Must benchmark target browser/device
Drift, obfuscation, monitoring	NOT COMPLETE	Needs recurring updates and adversarial tests
Context beyond URL text	LIMITATION	DNS/TLS/domain age/page signals recommended

Verdict: HYB-MS +10% is the strongest deployment candidate in this experiment. It is not yet deployment-ready.